Article Original **Open Access**

Hepatitis C infection markers among blood donors in Kisangani, Democratic Republic of Congo

BATINA AGASA S[*], DUPONT E[**], KAYEMBE T[*], BOLUKAOTO B[*], KAMBALE K[*], TSHOMBA O[*], MALENGELA R[***], MBONGO C[***], VERTONGEN F[****], GULBIS B[***]

Affiliation: [*]Cliniques Universitaires, Université de Kisangani, [**]Laboratoire d'Immunologie-Hématologie-Transfusion [***]Centre Provincial de Transfusion Sanguine, République Démocratique du Congo et [****]Laboratoire de Chimie Médicale, Hôpital Erasme, Université Libre de Bruxelles, Bruxelles, Belgique.

Address for correspondence: Batina Agasa, Internal Medicine department, School of Medicine, University of Kisangani BP. 2012 Kisangani /Congo/RDC. Av. Munyororo, Kisangani, Republic Democratic of Congo Cell phone.: +243 85 384 66 80; e-mail: drbatinaagasa@hotmail.com (main adresse)

Cite this article: Batina As, Dupont E, Kayembe T et al**. Hepatitis C infection markers among blood donors in Kisangani, Democratic Republic of Congo.** *Rev. méd. Gd. Lacs* 2012, **1**(4): 212-220

Article avalable at :

http://www.grandlacs-medjournal.com/images/stories/Hepatitis_C_infection_markers_among_blood_donors_in_Kisangani.pdf

http://www.grandlacs-med-journal.com/images/stories/vol_1Num_2.pdf/2012.pdf

Status of the article : peer reviewed

Batina AS et al. ©Revue Médicale de Grands Lacs - eISSN : 2304-0173 Il s'agit d'un article en accès libre distribué sous les termes de la 'Creative Commons Attribution License ' (http://creativecommons.org/licenses/by/2.0), qui permet l'utilisation sans restriction, la distribution et la reproduction sur tout support, à condition que le travail original soit correctement cité.

E : ISSN: 2304-0173
P : ISSN: 2304-0149

RESUME

Objectifs

Le rôle de la transfusion sanguine dans la propagation du virus de l'hépatite C (VHC) est préoccupant en RDC. Dépisté depuis fin 2004 chez les donneurs de sang, peu des données sont cependant disponibles sur le VHC à Kisangani. Une étude s'avère indispensable afin de déterminer la séroprévalence du VHC chez les donneurs de sang.

Patients et méthode

1247 échantillons de sang prélevés chez tous les donneurs bénévoles ayant fait don de sang du 1er août 2005 au 30 avril 2006 au Centre Provincial de Transfusion Sanguine ont été analysés pour la recherche d'anticorps anti-VHC. En même temps que la sérologie anti-VHC, les marqueurs du VIH et du VHB étaient dépistés.

Résultats

Au total 51(4,1%) donneurs bénévoles de sang (tableau I) étaient porteurs des anticorps anti-VHC. Cinquante-deux (4,2%) des sujets étaient séropositifs au VIH et 60 (4.8%) au VHB. L'âge moyen des donneurs séropositifs au VHC est de 31,4 ans (±13,1) (tableau II). La sérologie positive au VHC est moindre chez les donneurs de 17 à 24 ans par rapport à ceux qui ont 25 ans et plus ($p < 0,05$). La sérologie positive au VHC n'est pas liée au sexe.

Conclusion

La séroprévalence de virus de l'hépatite C est relativement élevée comme celle du VIH chez les donneurs bénévoles de sang à Kisangani. Elle justifie que tout donneur de sang soit dépisté pour le VHC afin de prévenir sa transmission à Kisangani.

Mots-clés : Hepatitis C infection markers, Blood donors, Kisangani, Congo, infection, Hepatitis C, transfusion, Hepatitis C virus.

ABSTRACT

Background

The role played by blood transfusion in the spread of hepatitis C virus (HCV) in the Democratic Republic of Congo (DRC) is a concern. HCV was detected in late 2004 among blood donors in the DRC, and few data are available on HCV in Kisangani. Determining the sero-prevalence of HCV among blood donors is essential.

Patients and methods

1247 échantillons de sang prélevés chez tous les donneurs bénévoles ayant fait don de sang du 1er août 2005 au 30 avril 2006 au Centre Provincial de Transfusion Sanguine ont été analysés pour la recherche d'anticorps anti-VHC. En même temps que la sérologie anti-VHC, les marqueurs du VIH et du VHB étaient dépistés.

Results

In total, 51 (4.1%) voluntary blood donors were carriers of HCV antibodies. Fifty-two donors (4.2%) were HIV-positive and 60 (4.8%) were HBV-positive. The average age of HCV-positive donors was 31.4 years (±13.1). HCV-positive serology was significantly lower in donors between the ages of 17 and 24 years ($p<0.05$). Gender did not have an impact on HCV sero-prevalence.

Conclusion

The sero-prevalence of hepatitis C is relatively high, similar to that of HIV, in volunteer blood donors in Kisangani. This result demonstrates the need for mandatory HCV screening of all blood donors to prevent HCV transmission in Kisangani.

Keywords: Hepatitis C, transfusion, Hepatitis C virus, Hepatitis C infection markers, blood donors, Kisangani, Democratic Republic of Congo

Introduction

The introduction of serological screening for hepatitis C virus (HCV), introduced during the late 80s and early 90s in Europe and the USA, has decreased the risk of transmission via transfusion to less than 1:250,000. [1, 2] In resource-limited countries in Sub-Saharan Africa, the introduction of HCV testing of blood donors was delayed and has occurred at different times across the continent.

In the Democratic Republic of Congo (DRC), this test was introduced in Kisangani in late 2004, particularly in the the Provincial Blood Transfusion Centre (PBTC). This ensures that the samples were obtained from volunteer donors at the same centre or blood drive. The donors, aged 18 to 60 years, underwent further clinical examination, immuno-haematological examinations, and testing to ascertain the human immunodeficiency virus (HIV), hepatitis B virus (HBV), HCV, and syphilis status of the donor. Blood packets that were positive for any one of the markers were destroyed, while those negative for all markers were sent to appropriate hospitals. The average storage duration for packets was approximately one week due to high demand. Besides the CPTS, there are five integrated centres that supply hospitals, one of which is academic. These other centres accounted for nearly three-quarters of the supply, mainly from replacement donors recruited among relatives or close friends of the patient. Within health centres, which are facilities run by nurses, HIV testing was often the only test used to select the donor, testing for syphilis and HBV was done randomly, and HCV testing was not performed at that level.[3]

Few data on HCV are available in Sub-Saharan Africa in general and the DRC in particular. Previous studies have suggested a link between blood transfusion and HCV infection. An evaluation carried out in Kinshasa in 2001, for example, reported a relationship between HCV seropositivity and transfusion history among female sex workers. [4] Among patients with sickle cell disease that received blood transfusions, 8% were HCV-positive. [5] This figure may be an indication of the HCV infection frequency in blood donors, given the possible transmission routes of HCV.

Thus, determining the sero-prevalence of HCV among blood donors and proposing measures to improve safety in blood transfusion is necessary.

Materials and Methods

All 1247 volunteer blood donors who donated blood from August 1, 2005 to April 30, 2006 were examined for the presence of HCV antibodies. This anti-HCV test was combined with screening for markers of HBV and HIV. Serological markers for HIV and HBV were determined using Determine®TM VIH-1/2 combined with Capillus® and Oraquick® for HIV and DetermineTM® HBsAg (Abbott, Tokyo, Japan) for HBV. For HCV, the Hexagon® HCV test (HumanGmbH, Wiesbaden, Germany) was used.

For statistical analysis, the chi-squared test was used with a significance level of 0.05.

Results

Seroprevalence: In total, 51 (4.1%) volunteer blood donors (**Table I**) possessed HCV antibodies. Fifty-two subjects (4.2%) were HIV-positive and sixty (4.8%) were HBV-positive.

Table I. Seroprevalence of HCV, HIV and HBV among blood donors (see below**)**

Age and Sexe: The average age of HCV-positive donors was 31.4 years (±13.1; **Table II**). The sero-prevalence of HCV in donors aged between 17 to 24 years old was significantly lower than that of donors aged 25 years and older ($p<0.05$). However, positive serology for HCV was not related to gender.

Table II. Age and sexe of blood donors according to anti-HCV antibodies (see below)

Discussion

The sero-prevalence of 4.1% for HCV is comparable to that for HIV found in the same subject population (source). The role of blood transfusion in the transmission of HCV in the country has been discussed.[4] However, HCV infection is largely ignored in the face of HIV infection, [6] which has a very similar pattern of transmission routes.

HCV is considered endemic within Sub-Saharan Africa, although prevalence of HCV infection varies greatly between nations, from 8.09% in Benin [7] to 0.3% in Djibouti. [8] In most studies, the sero-prevalence in the general population remains around the value found in our study. For example, it was found to be 3.6% in the Central African Republic, 4.8% in Chad and Cameroon, and 5.7% in Angola.[9, 10] The outlying results in Sub-Saharan Africa may be due to other categories of donors included in the study. In Brazzaville, the overall sero-prevalence was found to be 3.9%, 0.37% among volunteer donors unrelated to the patient and 6.71% among family donors. [9] Furthermore, in Mali, HCV sero-prevalence was found to be 4.5% among donors giving their first blood donation and 0.8% among regular donors. [11]

In this study, HCV seropositivity was observed in almost all age groups, which illustrates the endemic character of HCV in the region. However, the sero-prevalence of HCV among those aged 25 years and older was found to be higher than among younger donors (17 to 24 years of age). The HIV infection rate is also higher among the former than in the latter. Similar observations have been reported in other countries in the region, such as Namibia. [12]

The subjects under 25 years of age currently represent the largest category of the population in the DRC, [13] which presents an opportunity. This population should be encouraged to donate blood. Thus, the number of volunteer donors at low risk would increase, leading to an increase in blood safety and the supply of blood at the same time. [14-16]

The medical history taken before blood donation leads to the exclusion of potential donors that have

engaged in risky sexual behaviour or drug abuse, those with a history of jaundice, and those with a history of transfusion. In our setting, procedures that have been performed with non-disposable equipment, such as circumcision, tattooing, piercing, home births, etc., represent potential risk factors that require extensive evaluation.

Conclusion

The sero-prevalence of HCV is comparable to that of HIV in blood donors in Kisangani. This result demonstrates that screening for HCV in all blood donors in hospitals and others facilities should be mandatory.

References

1. Allain, JP. **Transfusion risks of yesterday and of today.** *Transfus Clin Biol* 2003; **10**: 1-5.

2. Dodd R, Kurt Roth W, Ashford P, et al. **Transfusion medicine and safety.** *Biologicals* 2009; **37**: 62-70.

3. Centre provincial de transfusion sanguine (CPTS) Annual Activity Report, Kisangani. 2006.

4. Laurent C, Henzel D, Mulanga-Kabeya C, et al. **Seroepidemiological survey of hepatitis C virus among commercial sex workers and pregnant women in Kinshasa, Democratic Republic of Congo.** *Int J Epidemiol* 2001; **30**:872-7.

5. Batina AS, Dupont E, Kayembe T, et al. **Multiple transfusions for sickle cell disease in the Democratic Republic of Congo: The importance of the hepatitis C virus.** *Transfus Clin Biol* 2010; **17**: 254-9.

6. Sodahlon YK, Segbena AY, Prince-David M, et al. **[Blood transfusion safety in a limited resources setting: The elaboration of a National Blood Policy in Togo].** Article in French. *Santé* 2004;**14**: 115-20.

7. Abdoulaye I, Siebertz B, Akpona S, et al. **[Seroprevalence of anti-hepatitis C virus among blood donors in Benin].** Article in French.*Gazette de la transfusion* 2005: **191**.

8. Dray X, Dray-Spira R, Bronstein JA, Mattera D. **[Prevalences of HIV, hepatitis B and hepatitis C in blood donors in the Republic of Djibouti].** Article in French. *Med Trop* (Mars) 2005; **65**:39-42.

9. Elira Dokelias-A, Okandze-Elenga JP, Dzia-LepfounzouA, et al **Prévalence des marqueurs viraux majeurs chez les donneurs de sang à Brazzaville.** *La Gazette de la Transfusion* 2002; **177**:4-176.

10. Mbanya DN, Takam D,Ndumbe PM. **Serological Findings amongst first-time blood donors in Yaounde, Cameroon: Is safe donation a reality or a myth?** *Transfus Med* 2003; **13**: 267-73.

11. Diarra A, Kouriba B, Baby M, et al. HIV, HCV, HBV and **syphilis rate of positive donations among blood donations in Mali: Lower rates among volunteer blood donors.** *Transfus Clin Biol* 2009;**16**:444-7.

12. Vardas E, Sitas F, Seidel K, et al. **Prevalence of hepatitis C virus antibodies and genotypes in asymptomatic, first-time blood donors in Namibia.** *Bull World Health Organ* 1999; **77:**965-72.

13. World Bank.Democratic Republic of Congo Health, Nutrition and Population Country Status Report.2005: Report no. 35626-ZR.

14. Jones RL. **Young donors:Opportunity, responsibility, vulnerability –report onYDAR task force.** *Transfusion* 2009; **49**: 2019-20.

15. No authors listed. (2010). **Young blood wanted**. *Lancet* 2010; **375**: 2122.

16. Davey RJ. **Recruiting blood donors: Challenges and opportunities.** *Transfusion* 2004; **44**: 597-600.

Table I. Seroprevalence of HCV, HIV and HBV among blood donors

Viral markers	N	%
HCV	51	4,1
HIV	52	4,2
HBV	60	4,8
VHC + VIH	4	0,3
VHC + VHB	5	0,4

VHC: Hepatitis virus C, VHB: Hepatitis virus B

Hiv: Human immunodeficiency virus

Table II. Anti-HCV antibodies by age ranges and sex

	HCV		Total	p
	Positif	Negatif		
Age (years)				
Mean (SD)	31,4 (13,1)	27,3 (10,6)	27,49(10,8)	
17 - 24	12	749	761	< 0.01
25 - 32	25	277	302	
33 - 40	9	84	93	
41 - 48	4	58	62	
49 - 56	1	25	26	
57 - 64	0	3	3	
Sexe				
F	11	251	262	NS
M	40	945	985	

NS: non-significant